About the Cover

The cover is an equity implied volatility surface, the object of Chapter 19 and the one picture that says what this book is about.

The curves running across the sheet are smiles, one to a maturity: implied volatility against moneyness, out-of-the-money puts to the left, calls to the right. The lines running the other way join them along expiration, so the surface reads as a mesh rather than as a set of separate curves. Maturity runs back to front, two years at the back and six weeks at the front, and read in that direction you watch the skew steepen as expiration approaches: the most durable empirical regularity in equity option markets, and the one thing a single Black–Scholes volatility cannot accommodate, since it would draw this picture as a flat plane. The colors run with maturity, the amber of the near smiles cooling to blue at the long end; they are the two colors the interior uses for its worked examples and its case studies.

The shape is not artistic license. It is drawn from

$$\sigma(k,T) \;=\; 0.18 \;+\; \frac{0.11\,k^2}{T+0.30} \;-\; \frac{0.075\,k}{T+0.50},$$

with k the log-moneyness and T the maturity in years: a convexity term and a skew term, each decaying with maturity at its own rate. Curvature and tilt therefore fade at different speeds, which is what makes the surface twist as it flattens rather than simply subside.

That it is computed rather than drawn is the point, and it is the rule the whole book follows. Every plot here is generated in TEX from the formula the text derives, so no curve can drift out of step with the equation beside it; the cover is held to the same standard as the figures inside it.

The surface is also the natural emblem for the title. It is a market observable, quoted and traded in its own right. It is the output of a model, and the place where models are most visibly asked to do more than they can. And its shape is a fact about market structure: who buys crash protection, and who is left short it. Contracts, models, and markets meet here, which is why this is the picture on the front.

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